Introduction

Queue using a circular list using an array
Data Structures and its Applications (UE25CS52A)

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Quiz : Queue

1. In our implementation, on initialization

a) Front and rear should be -1

b) Front and rear should be 0

c) Front and rear could be any valid index

d) None of these

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Quiz : Queue

2. On deque, queue is empty if

a) Front and rear are equal after incrementing front

b) Front and rear are equal after incrementing rear

c) Front and rear are equal before incrementing rear

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Quiz : Queue

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Quiz : Queue

3. deque need not consider this as a special case

a) Empty queue

b) Queue with one node

c) Queue with number of nodes

d) None of these

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Quiz : Queue

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Quiz : Queue

4. To support different type of component in the queue,

a) The type of front should change

b) The type of rear should change

c) The type of key should change

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Quiz : Queue

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Quiz : Queue

5. Deinitialize the queue

a) Requires number of operations as the number of components in the queue

b) Requires a few operations; does not depend on the number of components in the queue

c) Requires a looping structure

d) None of these

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Quiz : Queue

5. Deinitialize the queue

a) Requires number of operations as the number of components in the queue

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queue: structure

```
#define MAXSIZE 5
struct queue
{
        int key_[MAXSIZE];
        int front_;
        int rear_;
};
typedef struct queue queue_t;
```


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Queue: implementation

- Initialize / Deinitialize the queue
  - Initialize : make head and tail 0
  - Deinitialize : make head and tail 0
- Enqueue
  - Add at the rear
  - Increment and check whether queue is full
  - First node => modify front
- Deque
  - Check whether the queue is empty
  - If not increment front
  - Remove the front
  - Modify front

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Queue: implementation

- Initialize / Deinitialize the queue
  - Initialize : make head and tail 0
  - Deinitialize : make head and tail 0

```
void init_queue(queue_t *ptr_queue)
{
        // could be any valid index in the array
        ptr_queue->rear_ = ptr_queue->front_ = 0;
}
void deinit_queue(queue_t *ptr_queue) // NO LOOP!
{
        ptr_queue->rear_ = ptr_queue->front_ = 0;
}
```


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Queue: implementation

- is_full and is_empty : same code!!

```
// to be called before deque before incrementing
int is_empty(queue_t *ptr_queue)
{
        return ptr_queue->front_ == ptr_queue->rear_;
}

// to be called in enque after incrementing
int is_full(queue_t *ptr_queue)
{
        return ptr_queue->front_ == ptr_queue->rear_;
}
```


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Queue: implementation : **enque and deque**

```
void enque(queue_t *ptr_queue, int key)
{
        int temp;
        temp = ++ptr_queue->rear_ % MAXSIZE;
        ptr_queue->rear_ = temp;
        if(! is_full(ptr_queue))  {
                ptr_queue->key_[temp] = key;
        }
        else {
                ptr_queue->rear_ = (ptr_queue->rear_ - 1) % MAXSIZE;
                printf("queue full; cannot push\n");
        }
}
```


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Queue: implementation : enque and deque

```
int deque(queue_t *ptr_queue)
{
        if(! is_empty(ptr_queue)) {
                int key;  int temp = ++ptr_queue->front_ % MAXSIZE;
                key = ptr_queue->key_[temp];
                ptr_queue->front_ = temp;
                return key;
        } else {
                printf("queue empty; cannot pop\n");
                return '\0';
        }
}
```


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Queue: Applications

- CPU scheduling : processes in batch processing usually wait in a queue
- Printer Spooler : collects the jobs for printing in a queue
- Breadth First search : Looking for a key in a tree structure
- Simulation models : queues to simulate random events in queueing theory
- Message queues : sharing information in operating systems
- Toll booth : waiting of vehicles to pass through a toll booth

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THANK YOU

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